Chapter 1: Introduction and Overview of GIS and Software [3 Hours]

1.1 Definition of a GIS

- **Theory Questions**

1. Define GIS. [Part of 3 (Probably 1)] (2076 Bhadra, Q1a)
2. Write down the definition of GIS. [Part of 10] (2074 Magh, Q1)
3. Define Geographic Information System with figure. [2] (2072 Magh, Q1)
4. Write about different components of GIS [4] (2079 Chaitra, Q1a)
5. Explain the functional components of a GIS. [4] (2071 Magh, Q1a)
6. Write down the definition of GIS, its components, and basic types of spatial data [6] (2071 Bhadra, Q1a)
7. Write down the definition of GIS, its components, and basic types of spatial data [Not specified] (2075 Magh, Q1a)
8. What is the difference between spatial and non-spatial data? [2] (2071 Bhadra, Q1b)
9. What is the difference between spatial and non-spatial data? [Not specified] (2075 Magh, Q1b)
10. • How does GIS work? What are the essential features of GIS? (2082 Shrawan)
11. • Define GIS and explain its key features and functions. (2081 Chaitra)

1.2 Features and Functions

- **Theory Questions**

1. Write down the major functions of GIS [2] (2071 Bhadra, Q1c)
2. Describe about four Ms. [Not specified] (2075 Bhadra, Q1b)
3. Which basic function of GIS is the most important? Why? [Not specified] (2075 Bhadra, Q1c)

1.3 Why GIS is Important

- **Theory Questions**

1. What is GIS and its importance in civil engineering? [2+4] (2073 Magh, Q1)
2. Discuss why GIS is an important tool in today's world... (2081 Chaitra)

1.4 How GIS is Applied

- **Theory Questions**

1. Write any four applications of GIS. [Not specified] (2072 Magh, Q1)
2. List some of the applications of GIS. [2] (2071 Magh, Q1a)
3. How GIS can be applied in infrastructure planning and design? [Not specified] (2075 Bhadra, Q1a)
4. provide at least three examples of how it is applied in different fields. (2081 Chaitra)

1.5 GIS as an Information System

- **Theory Questions**

No direct questions asked.

1.6 GIS and Cartography

- **Theory Questions**

1. Briefly explain the difference between GIS and cartography [5] (2072 Ashwin, Q1a)
2. Write short notes on: GIS and cartography [3] (2071 Magh, Q6e)
3. Differentiate GIS from cartography, CAD, and information system [3] (2073 Bhadra, Q1a)
4. Differentiate GIS from cartography and database management system [Part of 3 (Probably 2)] (2076 Bhadra, Q1a)
5. Explain the development of GIS from cartography, CAD, and Database management system [4] (2078 Chaitra, Q1a)

1.7 Contributing and Allied Disciplines

- **Theory Questions**

1. Explain the development of GIS from cartography, CAD and Database management system. (2078 Chaitra)

1.8 GIS Data Feed

- **Theory Questions**

1. Explain data feeds of GIS [3] (2073 Bhadra, Q1b)
2. Explain primary and secondary sources of GIS data. Describe some recent developments in data acquisition techniques. (2082 Shrawan)
3. What are the major data sources of GIS data? Why data collection is the most important process in GIS analysis? (2078 Chaitra)
4. What are the basic sources of data for GIS? (2076 Bhadra)
5. Write the sources of data feed to GIS and their characteristics. (2072 Ashwin)

1.9 Historical development of GIS

Theory Questions

1. (Covered in 1.6 2078 Chaitra, Q1a)

Chapter 2: GIS and Maps [3 Hours]

2.1 Map Projections and Coordinate Systems

- **Theory Questions**

1. Define Projection. [2] (2071 Bhadra, Q2a)
2. What do you understand by Map projection system? [2] (2073 Bhadra, Q2a)
3. Why is datum important in GIS? Describe about two common datums? Describe in brief three types of map projections [1+2+] (2072 Magh, Q2)
4. Why is map projection important? Explain which map projection system is used for preparing maps of Nepal? [2+3] (2072 Ashwin, Q1b)
5. What is map projection? List the three main categories of map projection. Explain Nepalese Projection System with all parameters. [1+1+4] (2071 Magh, Q1b)
6. Describe briefly the projection system which has been used in Nepal? [Not specified] (2075 Bhadra, Q2a)
7. Explain Nepal datum and projection system [2] (2071 Bhadra, Q2b)
8. Describe Universal Transverse Mercator and Modified Universal Transverse Mercator projection system for Nepal and write down the parameters used in these projection systems [5] (2074 Magh, Q2a)
9. Why is map projection necessary? State two main differences between a Geographic Coordinate System and Projected Coordinate System [4] (2073 Magh, Q2a)
10. Write about the parameters of coordinate system used in Nepal [2] (2073 Bhadra, Q2b)
11. Write the differences between UTM coordinate system and coordinate system(s) used in Nepal? [3] (2076 Bhadra, Q2a)
12. Why Nepal uses a different coordinate system than the UTM? [2] (2073 Magh, Q2b)
13. Why Nepal's Coordinate system (MUTM) uses a 3° zone instead of 6° compared to UTM? [2] (2073 Bhadra, Q2c)
14. Explain the relationship between datums, projections, and coordinate systems in GIS. What are the UTM zones used in Nepal and which areas are covered by each of the zones? [4+2] (2080 Chaitra, Q2)
15. Explain Nepal and UTM coordinate system [Not specified] (2075 Magh, Q1c)

16. Why civil engineering projects requires to use projected coordinate system for the analysis? [3] (2078 Chaitra, Q2b)
17. Write short notes on: Coordinate system [3] (2071 Magh, Q6c)
18. Write short notes on: UTM and MUTM coordinate system [2] (2074 Bhadra, Q6iv)
19. Explain the concept of map projections and why they are necessary. Discuss the different types of map projections and their advantages and disadvantages. (2081 Chaitra)
20. In a Geographic Coordinate System, what is the significance of the Prime Meridian and the Equator? (2082 Shrawan)

2.2 Maps and Their Characteristics (Selection, Abstraction, Scale, etc.)

- **Theory Questions**
  1. What is scale and write down different types of scales with example? [6] (2071 Bhadra, Q10)
  2. Map is a 2D model of reality. Explain this with examples. (2082 Shrawan)

2.3 Automated Cartography versus GIS

- **Theory Questions**
  - No direct questions asked.

2.4 Map Projections

- **Theory Questions**
  - Covered in 2.1

2.5 Coordinate Systems

- **Theory Questions**
  - Covered in 2.1

2.6 Precision and Error

- **Theory Questions**

1. Explain about accuracy and precision [Part of 3] (2076 Bhadra, Q2b)
2. Explain accuracy and precision in GIS [Part of 5] (2074 Magh, Q2b)
3. Describe briefly on errors in GIS data made through actual changes [Not specified] (2075 Bhadra, Q2c)

- **Numerical Questions**

1. What is the allowable RMS error for 1:20000 scale map? [3] (2079 Chaitra, Q2a)
2. Find the distance between points A(24.5°N, 83.5°E) and B(24.6°N, 84.5°E) in km, for the radius of the earth 6378 km assuming earth's shape as a sphere [3] (2079 Chaitra, Q2b)

3. What is the allowable RMS error for 1:25,000 scale map? [3] (2078 Chaitra, Q2a)
4. What is the allowable RMS error for 1:25,000 scale map? [Part of 3] (2076 Bhadra, Q2b)
5. What would be the difference between the distance on earth presented by 1-degree latitude and 1-degree longitude if the radius of the earth at the equator is 6378137 m and the radius of earth along poles is 6356752.3142 m for WGS1984 system? [Not specified] (2076 Bhadra, Q2c)
6. Calculate allowable RMS error for a map of scale at 1:2500 and is in UTM projection. 1.29 m [Not specified] (2075 Bhadra, Q2b)
7. Calculate the allowable errors for 1:30,000 and 1:5,000 scale maps [Part of 5] (2074 Magh, Q2b)
8. Calculate the allowable RMS error in 1:15,000 & 1:5,000 map [2] (2071 Bhadra, Q2c)

Chapter 3: Spatial Data Models [3 Hours]

3.1 Concept of Data Model

- **Theory Questions**

1. Explain different types of data models [4] (2073 Bhadra, Q3a)
2. Explain different types of data models [3] (2076 Bhadra, Q3a)
3. Explain different types of data models in GIS. [6] (2078 Chaitra, Q1)
4. What is the Network data model? Describe with examples [Not specified] (2075 Bhadra, Q5)
5. Differentiate between qualitative and quantitative attributes in a vector dataset. [4] (2080 Chaitra, Q8a)
6. Write short notes on: Different types of attribute variable used in GIS [2] (2074 Bhadra, Q6iii)
7. What are the levels of measurement of attribute data? How do they affect the symbology? [6] (2079 Chaitra, Q3a)
8. How different symbologies does differ according to scaling of data (Nominal, ordinal, interval, and ratio)
9. Explain four levels (nominal, ordinal, interval, and ratio data) of commonly recognized data measurement. (2081 Chaitra)

3.2 Raster Data Model

- **Theory Questions**

1. Explain Vector model and Raster GIS Model with suitable examples [Part of 10] (2074 Magh, Q1)
2. How Raster and Vector model are differentiated? [2] (2072 Ashwin, Q1c)
3. Discuss the key differences between vector and raster data models, highlighting their strengths and weaknesses in various GIS applications. Include a sketch to visually represent how the same geographic features (a building, a road, and a few trees) might be depicted in each model [3+4] (2080 Chaitra, Q1)
4. Explain different types of Raster Models [6] (2071 Bhadra, Q3a)
5. What are the basic elements of raster data model? Explain the advantage and disadvantage of the raster data model versus vector data model [3+] (2072 Magh, Q3)

3.3 Compression

- **Theory Questions**

1. Explain the raster compression techniques with appropriate examples [4] (2079 Chaitra, Q4b)
2. Define the terms region quadtree and vectorization. [1+1] (2071 Magh, Q2a)
3. Explain run length coding and quadtree raster compression techniques with clear illustration [6] (2079 Chaitra, Q3b)
4. Explain run length and quad tree compression technique with clear illustrations. [6] (2078 Chaitra, Q3b)
5. Illustrate BSQ, BIL, and BIP raster format [Not specified] (2075 Magh, Q5d)
6. Explain run length and quadtree compression techniques for raster data. (2081 Chaitra)

- **Numerical Questions**

1. Explain run length encoding using the following raster data. Also produce each of the following encoding for this data:

   (a) Quad tree encoding

   (b) Block encoding

   (c) Chain encoding

25	25	31	31	25	25	3	3
25	25	31	31	25	25	3	3
25	25	31	31	25	25	3	3
25	25	31	31	25	25	3	3
27	27	27	27	27	27	27	27
27	27	27	27	27	27	27	27
27	27	27	27	27	27	35	27
27	27	27	27	27	27	27	27

[4+2+2+2] (2080 Chaitra, Q7)

2. How will the following land use raster be compressed using quadtree compression technique and run length coding technique? [6] (2076 Bhadra, Q3b)

1	1	1	1	3	3	4	4
1	1	1	1	3	3	4	4
1	1	1	1	3	3	4	4
1	1	1	1	5	5	5	4
2	2	2	2	5	5	5	5
2	2	2	2	5	5	5	5
1	1	1	1	3	3	5	5
1	1	1	1	3	3	5	5

3. Compress the given raster data using the quadtree method [Not specified] (2075 Bhadra, Q3a)

12	12	12	12	10	10	10	10
12	12	12	12	10	10	10	10
12	12	12	12	10	10	10	10
12	12	12	12	10	10	10	10
9	9	7	2	6	6	4	4
9	9	1	3	6	6	4	4
3	3	8	8	2	2	5	5
3	3	8	8	2	2	2	5

4. Write down the three raster encoding methods and % decrease in size for the following figure [6] (2074 Bhadra, Q1b)

25	25	25	25	25	25
25	25	25	25	25	25
25	25	25	25	25	25
5	5	5	10	10	10
5	5	5	12	12	12
5	5	5	15	12	12

5. How is the following raster encoded using full raster encoding, run length encoding, quad tree encoding, and value point encoding? [6] (2073 Magh, Q3)

F	F	F	F	W	W	F	F
F	F	F	F	W	W	F	F
F	F	F	F	O	O	F	F
F	F	F	F	O	O	F	F
F	F	F	F	S	S	S	S
F	F	F	F	S	S	S	S
F	F	F	F	S	S	W	W
F	F	F	F	S	S	W	W

*F= Forest
*W= Water Body
*O= Open Space
*S= Settlement

6. Show the encoding of the following raster using run length and Quadtree encoding [6] (2073 Bhadra, Q3c)

12	12	12	12	12	12	12	12
12	12	12	12	12	12	12	12
12	12	12	12	12	12	12	12
12	12	12	12	12	12	12	12
5	5	6	6	9	9	9	9
5	5	61	62	9	6	9	9
7	7	7	7	9	9	9	9
7	7	7	7	9	9	9	9

7. Explain run-length and Quad tree compression techniques. What would be the result of compressing the following data using both of the above techniques?

88	88	75	34	9	9	9	9
88	88	33	75	9	9	9	9
88	88	88	34	9	9	9	9
77	77	77	77	9	9	9	9
56	56	56	56	56	56	10	10
56	56	56	56	33	25	10	10
56	56	56	56	33	33	33	9
56	56	56	56	33	33	33	33

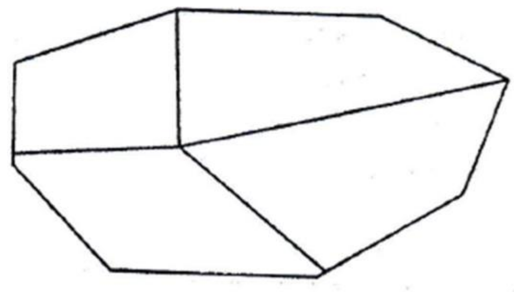
3.4 Indexing and Hierarchical Structures

- **Theory Questions**
  - No direct questions asked.

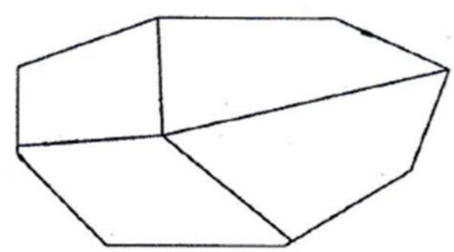
3.5 Vector Data Model

- **Theory Questions**

1. Label vertex, node, arc, and polygon for the following figure [2] (2074 Bhadra, Q1c)



2. Label vertex, node, arc, and polygon for the given figure and write different vector models of the given figure [6] (2071 Bhadra, Q3b)



3.6 Topology

- **Theory Questions**

1. Why is topology important in GIS? [Not specified] (2075 Bhadra, Q3b)
2. Write short notes on: Topology [3] (2071 Magh, Q6b)
3. What is topology in TIN data model? [2] (2073 Bhadra, Q3b)

3.7 TIN Data Model

- **Theory Questions**

1. Explain TIN Data model. [4] (2072 Ashwin, Q1c)
2. Does TIN belong to a raster or a vector and why? [4] (2071 Magh, Q2a)

Chapter 4: Data Sources [3 Hours]

4.1 Data Input and Data Quality

- **Theory Questions**

1. What are the primary data source for digitizing? Explain the difference between location errors and topological errors [2+4] (2072 Magh, Q6)

2. If a surveyor collects data using GPS, how can this data be integrated into a GIS system for mapping and spatial analysis? What steps are involved? (2081 Chaitra)

4.2 Major Data Feeds to GIS and Their Characteristics

- **Theory Questions**

1. Explain the major data sources for Geographic Information System [6] (2072 Magh, Q4)
2. Write the sources of data feed to GIS and their characteristics [5] (2072 Ashwin, Q2a)
3. What are the main sources of data feed to GIS? What is positional and attribute accuracy? [4+2] (2073 Magh, Q4)

4.3 Maps, GPS, Images, Databases, Commercial Data

- **Theory Questions**

1. What are the basic sources of data for GIS? [3] (2076 Bhadra, Q1b)
2. Explain the data sources of GIS. Why acquisition of data is considered an important task in GIS? [2+2] (2079 Chaitra, Q1b)
3. What are the major data sources of GIS data? Why data collection is the most important process in GIS analysis? [4] (2078 Chaitra, Q1b)

4.4 Locating and Evaluating Data

- **Theory Questions**
  - No direct questions asked.

4.5 Data Formats, Data Quality, Metadata

- **Theory Questions**

1. What are the direct and indirect methods of spatial data capture? List them. And explain any three data quality parameters [3+3] (2071 Magh, Q2b)

2. Describe briefly about the macro-level components of data quality [Not specified] (2075 Bhadra, Q4)
3. Write short notes on: Metadata [3] (2071 Magh, Q6a)
4. Explain the accuracy of GIS Data in terms of positional, logical, attribute, temporal accuracy and data completeness. What are the significance of data accuracy in GIS analysis? (2082 Shrawan)

Chapter 5: Database Concepts [3 Hours]

6.1 Database Concepts and Components

- **Theory Questions**

1. What is a database? What are the reasons for using a database management system? List out the four major types of database models used in GIS [1+3+2] (2071 Magh, Q3a)

6.2 Flat Files

- **Theory Questions**
  - No direct questions asked.

6.3 Relational Database Systems

- **Theory Questions**

1. Explain different types of database management system [3] (2071 Bhadra, Q4a)
2. Explain different types of database management systems [Not specified] (2075 Magh, Q4a)
3. Explain Rational data base system used in GIS [6] (2072 Ashwin, Q2b)
4. Explain hierarchical database and relational database model with figure [6] (2072 Magh, Q5)
5. What is a spatial DBMS? Explain different types of DBMS structures [2+3] (2073 Magh, Q5)
6. Explain: Relational Database Management System [4] (2074 Magh, Q4a)
7. Describe about RDBMS? Why is it used by GIS instead of other DBMS? [3] (2076 Bhadra, Q5a)
8. Why GIS uses relational database management system? [3] (2079 Chaitra, Q5a)
9. Explain RDBMS. How is the RDBMS used in GIS? [3] (2078 Chaitra, Q5a)
10. Explain about relational database management system [3] (2073 Bhadra, Q5a)
11. How relationships maintain in Relational Database Management System? [3] (2071 Bhadra, Q4b)
12. How the relationship is maintained? [3] (2073 Bhadra, Q5b)
13. How relationship maintains in relational database management system? [Not specified] (2075 Magh, Q4b)
14. Explain the concept of RDBMS (Relational Database Management System). How can you join two different tables having one common attribute using a SQL, assuming the following tables with common attribute landuse id in Table-1 and id in Table-2?
    Table-1: Vector data of landuse polygons with columns fid, landuse_id both being integer.
    Table-2: Non-spatial table describing the land use with columns id and, landuse type (such as

built-up, farmland etc.). (2082 Shrawan)

15. Write short notes on: Different types of database management system [2] (2074 Bhadra, Q6i)

6.4 Data Modeling

- **Theory Questions**
  - No direct questions asked.

6.5 Views of the Database

- **Theory Questions**
  - No direct questions asked.

6.6 Normalization

- **Theory Questions**

1. What is data Normalization? Explain about 1NF and 2NF [3] (2076 Bhadra, Q5b)
2. What is data normalization? Explain 1NF and 2NF [4] (2078 Chaitra, Q5b)

6.7 Database Analysis

- **Theory Questions**

1. What is Structured Query Language (SQL)? Give an example to select few rows of a database table based on a condition [3] (2080 Chaitra, Q9a)
2. What is the function of primary key in a table of a database? [3] (2080 Chaitra, Q9b)
3. What is meant by querying a database? How can you use a SQL to select all the rural municipalities having population between 10000 and 30000, provided that you have a polygon vector data of municipalities with columns: id, name, type-of-municipality (gaunpalika: rural municipality, nagarpalika, upmahanagarpalika, mahanagarpalika) and population? (2081 Chaitra)

Chapter 6: Vector Analysis [3 Hours]

7.1 Data Management Functions

- **Theory Questions**

1. Differentiate between data management functions and data analysis functions in vector GIS [5] (2072 Ashwin, Q4a)
2. Explain any three vector data management functions [6] (2072 Ashwin, Q4b)
3. Explain following vector function in vector GIS with suitable example

   (a) Erase

   (b) Clip

   (c) Map join

   (d) Update

   (e) Split

   [10] (2072 Magh, Q8)
4. Explain following vector function in vector GIS with suitable example

   (a) Union

   (b) Clip

   (c) Merge

   (d) Intersect

   [10] (2073 Magh, Q6)
5. Explain following vector functions with a suitable example (Spatial & Tabular):

   (a) Intersection

   (b) Clip

   (c) Dissolve

   (d) Union [10] (2071 Bhadra, Q5)
6. Explain following Geoprocessing tools with suitable spatial map and tables

   (a) Clip

   (b) Merge

   (c) Union

   (d) Intersection

   (e) Prove that Intersection = Union + Clip [12] (2074 Magh, Q3)

7. Explain clip, merge, union, dissolve, and intersection functions in vector GIS with suitable figures and tables for each function [16] (2074 Bhadra, Q4)

8. Explain in detail with appropriate example/illustration about following geoprocessing tools

   (a) Intersection

   (b) Union

   (c) Dissolve

   (d) Buffer

   (e) Clip

[10] (2079 Chaitra, Q6)

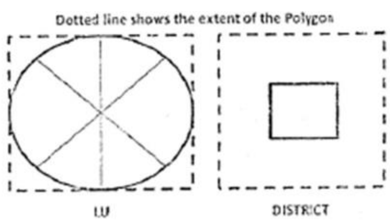
9. Differentiate between the following geoprocessing functions by suitable examples with features attribute tables

   (a) Clip and intersect

   (b) Merge and dissolve

[Not specified] (2075 Bhadra, Q6a, Q6b)

10. Write down the union, intersection, and clip functions (Figure and Table) from two polygons given below [Not specified] (2075 Magh, Q2a)



11. Dissolve the Lu table according to Lu_id and Name. [Not specified] (2075 Magh, Q2b)

Table of Lu polygon theme

Record_no	LU_id	name
1	100	Cultivation
2	200	Forest
3	200	Forest
4	100	Cultivation
5	100	Cultivation
6	300	Urban

Table of district polygon theme

Record_no	dis_id	D_name
1	20	Jumla

12. Explain buffer function with table [Not specified] (2075 Magh, Q2c)

13. Explain the following vector functions with appropriate examples including effects/changes in the attribute table: (i) Buffer (ii) Dissolve (iii) Union (iv) Intersection. (2082 Shrawan)

7.2 Data Analysis Functions

- **Theory Questions**

1. Define the term spatial analysis. Explain different types of vector overlay operations in GIS. Also explain three examples of neighborhood functions [2+4+4] (2071 Magh, Q7)
2. Write about the necessary tools and explain how will you perform following analyses using GIS? Describe the analysis tools along with relevant figures:

   (a) best location for landfill area so that following constraints are fulfilled.

Feature	Distance
Houses	> 1000 m
River	> 1500 m
Road	< 500 m
Water bodies	> 1500 m

   Provided data:

   Shapefile of river, land use data, point data of the houses, shapefile of roads, polygon shapefile of the study area.

   [10] (2078 Chaitra, Q6)

3. Write about the necessary tools and explain how will you perform the vector analysis using GIS to locate the best site for the dumping site area so that the following constraints are fulfilled? Describe the analysis tools along with relevant figures and attribute table data

Feature	Distance from
Settelement	>1500m.
River	>1000m.
Road	< 400m.
Water bodies	>2000m.

   Available data for analysis: Shapefiles of the river, land use data, point data representing location of the settlements, shapefile roads, polygon shapefile of the study area

   [10] (2076 Bhadra, Q6)

4. How will you perform following analyses using GIS? Describe the analysis tools along with relevant figures

   (a) Locate all the settlements that lies within 1 km distance (both side) from rivers

   (b) Flood plain zoning of the selected river according to types of river

   (c) Show the area of agricultural land that lies in that flood plain.

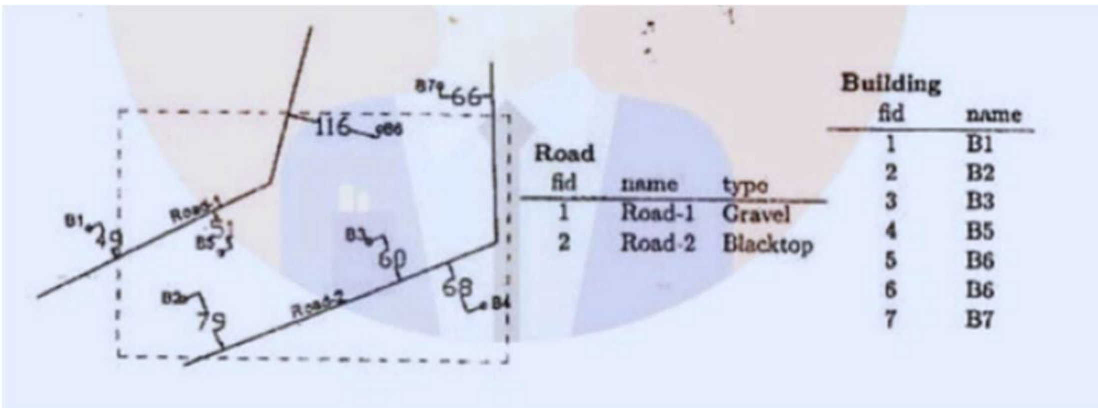
Provided data:

Shapefile of river, land use data, point data of the settlements, Shapefile of flood plain (Polygon)

[10] (2073 Bhadra, Q6)

- **Numerical Questions**

1. Figure below shows 2 roads, 7 buildings as points, and the area of interest (AOI) as polygon layers.



Road

fid	name	type
1	Road-1	Gravel
2	Road-2	Blacktop

Building

fid	name
1	B1
2	B2
3	B3
4	B5
5	B6
6	B6
7	B7

(a) Sketch the road and building layers after clipping with the AOI polygon (assume clip-road and clip-bld are the outputs respectively)

(b) Draw the 50 m buffer for each of the clipped layers along with clip-road and clip-bld layers (assume buffered layers are named as buf-road and buf-bld respectively)

(c) Sketch the intersection of buf-road and buf-bld and also show the output fields

(d) Sketch the union of buff-road and buf-bld and also show the output fields.

[2+2+3+3] (2080 Chaitra, Q4)

Chapter 7: Spatial Analysis [6 Hours]

7.1 Spatial Interpolation Methods

- **Theory Questions**

1. What is the purpose of spatial interpolation? Give a few examples where you would use spatial interpolation [4+2] (2079 Chaitra, Q5b)
2. Explain different types of interpolation in Raster GIS [Not specified] (2075 Magh, Q5a)
3. Explain different types of interpolation methods in raster GIS and their application in different engineering fields [4] (2074 Bhadra, Q3a)
4. Why is interpolation needed for the producing a DEM? Explain one such interpolation method [2+4] (2072 Magh, Q9)
5. What is spatial interpolation? Classify it. Compare the minimum curvature splines with inverse distance weighted interpolation method [2+3+5] (2071 Magh, Q9)
6. Explain the following with at least 1 relevant application example: (c) IDW interpolation [3] (2073 Magh, Q11c)
7. Explain the concept of spatial interpolation. Describe one method of spatial interpolation and discuss its advantages and disadvantages. (2081 Chaitra)

- **Numerical Questions**

1. Interpolate a value using any one method if the distance from four known values 160, 170, 180, and 90 are 1000, 400, 600, and 120 m respectively [4] (2074 Bhadra, Q3b) [Page 1]

7.2 Raster Analysis Including Topological Overlay

- **Theory Questions**

1. Differentiate between map overlay and map calculations. Explain how Raster analysis and topological overlay are carried out? [4+6] (2072 Ashwin, Q3a)
2. What are the advantages and disadvantages of raster based analysis? State four of the raster analysis operations in GIS. And explain them [3+3+4] (2071 Magh, Q8)
3. Explain proximity map calculation and summarize zone functions of Raster GIS [Not specified] (2075 Magh, Q5b)
4. How do you perform the following analyses using raster data? Illustrate with examples.

   i. Buffer

   ii. Intersection

   iii. Zonal statistics

   iv. Local operation such as mean,sum.etc.

[8] (2076 Bhadra, Q7a)

5. Explain and illustrate with examples the following raster functions
   (a) Slope and Aspect
   (b) Overlay
   (c) Zonal statistics
   (d) Hill shade
   (e) Euclidean Database

[10] (2079 Chaitra, Q7)

6. Explain and illustrate with examples the following raster functions.
   (a) Slope and Aspect
   (b) Cost path
   (c) Local and focal and zonal statistics
   (d) Hill shade and view shade
   (e) Euclidean Distance

[10] (2078 Chaitra, Q7)

7. Explain following raster spatial functions
   (a) Surface
   (b) Slope
   (c) Aspect
   (d) Viewshade
   (e) Map Algebra
   (f) Summarize Zone
   (g) Proximity
   (h) Distance

[16] (2074 Bhadra, Q5)

8. Explain the following Raster function with relevant example
   (a) View shade
   (b) Euclidean Distance
   (c) Reclassify
   (d) Slope and aspect
   (e) Hillshade

[10] (2073 Bhadra, Q7)

9. Explain the following with at least 1 relevant application example

   (a) Raster algebra

   (b) Zonal statistics

   (d) Reclassification

[3+3+3] (2073 Magh, Q11)

10. Explain following function in raster GIS with suitable example.

    (a) Map algebra

    (b) Slope

    (c) Hill shade

[4] (2072 Magh, Q7)

11. Explain following functions in Raster GIS with a suitable example:

    (a) Map Algebra

    (b) Interpolation

    (c) Slope

    (d) Aspect

    (e) Hillshade

[10] (2071 Bhadra, Q6)

12. Explain why calculation of NDVI from NIR and Red bands of satellite data is called local function in raster data analysis while calculation of slope and aspects are considered as neighborhood functions? [6] (2080 Chaitra, Q8b)

- **Numerical Questions**

1. Grid Code of a raster data with 25 m grid resolution is shown below. Derive a suitable map and suitable area for Rice Crop. Suitable parameters on 1 to 10 scale for slope, soil, and landuse are given in Table 1. The weightage for slope, soil, and landuse are 50%, 30%, and 20% respectively. Assume the value greater than 6 is suitable. [14] (2074 Magh, Q6)

1	1	1	2	2	3	4	5
2	2	2	3	3	3	4	4
1	1	2	5	5	5	5	4
3	5	5	4	4	4	4	4
3	5	5	4	4	4	4	4
1	3	3	3	3	3	4	4
1	2	2	2	1	1	1	1

Table 1: Suitable parameters

Gird Code	Suitable Parameters (1-10 scale)		
	Slope	Soil	Landuse
1	10	8	8
2	8	4	1
3	6	10	4
4	4	1	8
5	1	1	10

7.3 Map Calculations

- **Theory Questions**
  - (Covered under 7.2 in questions like 2071 Bhadra, Q6a and 2073 Magh, Q11a.)

7.4 Statistics

- **Theory Questions**

1. Describe briefly about cell statistics and zonal statistics [Not specified] (2075 Bhadra, Q7b)
2. (Also covered under 7.2 in questions like 2073 Magh, Q11b, 2078 Chaitra, Q7c, and 2079 Chaitra, Q7c.)

7.5 Integrated Spatial Analysis

- **Theory Questions**

1. You are working on a project to analyze the impact of deforestation on local water resources. What types of spatial data would you use, and how would you represent them in GIS? Describe with sketches [4] (2080 Chaitra, Q3)

2. You are tasked with identifying suitable locations for building a new reservoir in a region. The ideal location must meet the following criteria:
   (i) The area must have a slope of less than 10 degrees to ensure stability.
   (ii) It must be within 1 kilometer of a river to allow for water supply.
   (iii) The land cover must be non-forested (e.g., grassland or bare soil) to minimize environmental impact.
   (iv) The area must have a high water-retention capacity, indicated by soil types classified as "clay" or "silt."
   The data provided are DEM, Soil type, River, and Land cover data. Using raster data analysis, describe the steps you would take to identify suitable locations. Include at least four raster analysis functions in your explanation. (2081 Chaitra)

- **Numerical Questions**

1. Generate an output raster of the suitable area for landfill site from the given raster below. The values in both rasters given below denote the distances in kilometers from settlement buildings and river network. The area located farther from the buildings and closer from the river network is the suitable for it. First of all set the conditions for both raster. [Not specified] (2075 Bhadra,

4	8	3	3
3	1	5	5
6	4	11	12
9	5	14	15
4	8	16	17

Distance from buildings

11	10	9	9
10	9	8	7
6	5	2	1
5	4	3	2
5	4	3	2

Distance from river

Chapter 8: Surface Model [3 Hours]

8.1 DEM

- **Theory Questions**

1. What is a DTM? List the various structure of DTM. Explain three applications of DTM [1+2+3] (2071 Magh, Q3b)
2. Explain how DEM is created in Raster GIS? [5] (2072 Ashwin, Q2c)
3. What are the data sources of DEM? Describe about the applications of DEM [Not specified] (2075 Bhadra, Q8)

8.2 Slope

- **Theory Questions**

1. Explain about slope function with illustration. [Part of 5] (2076 Bhadra, Q7b)
2. (Also Covered under 7.2 in questions like 2071 Bhadra, Q6c, 2073 Bhadra, Q7d, 2078 Chaitra, Q7a, and 2079 Chaitra, Q7a.)

8.3 Aspect

- **Theory Questions**

1. Explain about aspect function with illustration. [Part of 5] (2076 Bhadra, Q7b)

2. (Also Covered under 7.2 in questions like 2071 Bhadra, Q6d, 2073 Bhadra, Q7d, 2078 Chaitra, Q7a, and 2079 Chaitra, Q7a.)

8.4 Other Raster Functions

- **Theory Questions**
  - (Covered in 7.2)

Chapter 9: River Network Generation [4 Hours]

9.1 Flow Direction

- **Theory Questions**

1. Define watershed and explain filled DEM and flow direction [1+5] (2072 Magh, Q10)
2. Explain Sink, Flow Direction, Flow Accumulation, River Grid, and stream lines [5] (2074 Magh, Q5a)
3. Explain how a sink is developed in DEM and how it can be removed [4] (2074 Bhadra, Q2a)
4. Explain how a sink is developed in DEM and how it can be removed [Not specified] (2075 Magh, Q3a)
5. Describe about the Shreve method for stream order [Not specified] (2075 Bhadra, Q9b)

- **Numerical Questions**

1. Following diagram shows an elevation raster data at 30 m spatial resolution. Refer to any cell as columns (A to F) and rows (1 to 6). For example the value of cell B2 is 118. Answer the questions based on this elevation raster.

	A	B	C	D	E	F
1	130	128	102	101	124	138
2	135	118	97	88	114	121
3	110	98	84	68	92	103
4	115	86	81	78	76	87
5	112	102	84	73	96	105
6	129	118	103	65	108	119

(a) Find the sink cell in the above elevation data. Determine the new elevation value to which each sink cell should be raised (fill sink) considering that the elevation should be at least 0.5m higher than the minimum elevation of the surrounding cell.

(b) Show the flow direction for all cells using arrows.

(c) Calculate flow accumulation for all cells.

(d) Considering the flow accumulation criteria for the formation of river as 4500 m², sketch the rivers

(e) Find the area of the biggest watershed basin from this data [2+2+2+3+3] (2080 Chaitra, Q10)

2. The following grid represents a DEM data of certain area whose cell size is 100 m. Using this data answer the following questions

78	72	69	71	58	49
74	67	56	49	46	50
69	40	44	37	37	48
64	58	55	22	22	24
78	61	47	21	21	19
74	53	34	12	11	12

(a) Show the sink cell

(b) What will be the value of that cell when the sink is filled?

(c) Calculate flow direction and flow accumulation value for each cell

(d) What is the area of the largest possible watershed?

(e) Show the river in the raster when stream definition is limited to minimum catchment area of 40000 m²

[10] (2079 Chaitra, Q8)

3. The following grid represents an SRTM DEM data with resolution 100 m.

1078	1072	1069	1071	1058	1049
1074	1067	1056	1049	1046	1050
1069	1040	1044	1037	1037	1048
1064	1058	1055	1022	1022	1024
1078	1061	1047	1021	1021	1019
1074	1053	1034	1012	1011	1012

(a) Find out the sink and fill it

(b) Show the flow direction Map

(c) Calculate and show the flow accumulation

(d) Delineate the watershed(s) and show the river when stream definition is limited to minimum 5 grids [2+2+2+2] (2078 Chaitra, Q8)

4. The following grid represents an Aster DEM data of certain area whose cell size is 30 m.

a) Write the flowchart of the watershed delineation

(b) Delineate the watershed(s) and show the river when stream definition is limited to minimum catchment area of 9000 m². Show the intermediate steps/calculations.

278	272	269	271	258	249
274	267	256	249	246	250
269	250	244	237	237	248
264	258	255	222	222	224
278	261	247	221	221	219
274	253	234	212	211	212

[8] (2076 Bhadra, Q8)

5. Derive flow direction, flow accumulation, and river network from the figure below. Consider threshold as 6 cells. If the cell size is 100 m, then calculate the catchment area at the lower right cell having the value 445. [Not specified] (2075 Magh, Q3b) [Page 2]

510	500	490	480	470
505	495	490	475	480
500	485	455	470	475
495	490	480	465	470
490	485	475	455	450
500	490	480	460	445

6. Derive flow direction from the following raster [Not specified] (2075 Bhadra, Q9a)

58	52	55	53	56	58
55	40	42	45	51	55
48	34	20	33	48	52
33	23	28	27	25	38
17	20	21	22	23	24
12	10	15	18	16	17

7. Derive flow direction, flow accumulation, river network, and catchment boundary using the following Digital Elevation Model and threshold value 4. [10] (2074 Magh, Q5b)

109	106	104	103	114	115
107	105	102	100	112	113
105	103	100	95	110	112
106	105	99	90	108	110
107	110	98	94	110	111
108	111	96	95	112	115

8. Derive flow direction, flow accumulation, and river network from the figure below. Consider threshold value as 10 cells. If the cell size is 100 m, then calculate the catchment area at the lower right cell having value 445 [12] (2074 Bhadra, Q2b)

480	500	490	480	485
505	495	490	475	480
500	485	455	470	475
495	490	480	465	470
490	485	475	455	450
500	490	480	460	445

9. Calculate the flow direction and flow accumulation raster from the following DEM. Also, delineate the watershed area if the pour-point reference cell is (4,4) and cell size is 30×30 meters [10] (2073 Magh, Q7)

0, 0

118	112	109	111	115
114	107	96	89	92
109	40	84	77	79
104	99	95	62	65
68	68	67	61	64
62	60	58	56	50

10. The following grid represents a DEM data of certain area whose cell size is 100 m. Using this data answer the following questions.

(a) Show the sink cell

(b) What will be the value of that cell when the sink is filled?

(c) Calculate flow direction and flow accumulation value for each cell

(d) What is the area of the largest possible watershed?

(e) Show the river in the raster when stream definition is limited to minimum catchment area of 40000 m²?

[12] (2073 Bhadra, Q8)

11. Generate flow direction, flow accumulation, and river network from the following DEM [8] (2072 Ashwin, Q5b)

100	98	97	98	99
105	103	102	99	100
107	105	95	100	101
109	106	104	102	104
115	112	110	109	108

12. Calculate the flow direction and flow accumulation raster from the following flow pattern of DTM in a given topography. Also derive river network grid if threshold value is 9 [2+2+2] (2071 Magh, Q4a)

156	144	138	142
148	134	112	98
138	106	88	74
128	116	110	44

13. Calculate the flow direction, flow accumulation, and river network grid from the following DEM grid. Generate river network with threshold value 5 [12] (2071 Bhadra, Q7)

93	90	89	90	89
91	88	82	79	78
89	71	72	73	74
86	83	82	70	69
86	84	82	75	68

78	72	69	71	58	49
74	67	56	49	46	50
69	40	44	37	37	48
64	58	55	22	22	24
78	61	47	21	21	19
74	53	34	12	11	12

125	123	121	120	118	117
124	118	119	117	116	118
123	120	118	116	114	115
122	118	116	115	113	114
121	118	114	112	112	114
120	117	115	113	109	112

14. The following grid represents a LiDAR-derived Digital Elevation Model (DEM) with a resolution of 10 meters. Use this data to perform the following tasks:
a) Identify and fill any sinks in the DEM. (This question includes a 6x6 grid with values 125, 123, 121...)
b) Delineate the watershed(s) and extract the stream network using a minimum catchment area of 1000 m².
Show all intermediate steps and calculations.

9.2 Flow Accumulation

- **Theory Questions**
  - (Covered under 9.1 in questions like 2074 Magh, Q5a.)
  - What are the steps involved in generating a river network within a given area (represented by a polygon vector data) assuming that multiple DEMs are available which covers the whole area? Explain steps with sketches. (2082 Shrawan)
- **Numerical Questions**
  - (Covered under 9.1 in numerical questions like 2071 Magh, Q4a, 2073 Bhadra, Q8, etc.)

9.3 River Network

- **Theory Questions**
  - (Covered under 9.1 in questions like 2074 Magh, Q5a.)
- **Numerical Questions**
  - (Covered under 9.1 in numerical questions like 2071 Magh, Q4a, 2073 Bhadra, Q8, etc.)

9.4 Watershed Delineation

- **Theory Questions**
  - (Covered under 9.1 in questions like 2072 Magh, Q10.)
- **Numerical Questions**
  - (Covered under 9.1 in numerical questions like 2073 Bhadra, Q8, 2078 Chaitra, Q8, etc.)

Chapter 10: GPS [4 Hours]

10.1 Basic Concept of GPS

- **Theory Questions**

1. Explain the components of a GPS [4] (2071 Magh, Q5a)
2. Explain about the components of GPS [3] (2076 Bhadra, Q4a)
3. Explain the functions of GPS segments [Not specified] (2075 Bhadra, Q10a)
4. Write short notes on: Advantages and disadvantages of GPS [3] (2071 Magh, Q6d)
5. Discuss the main components of a GPS system. Explain the roles of the space segment, control segment and user segment in GPS system. (2082 Shrawan)

10.2 How GPS Works

- **Theory Questions**

1. Explain Basic Principles of GNSS. [4] (2074 Magh, Q4b)
2. Explain basic principles and basic equation of GPS [2] (2071 Bhadra, Q8b)
3. Explain basic principles of GNSS and basic equation of GNSS [Not specified] (2075 Magh, Q4c)
4. Explain the guiding principles of GPS measurement with descriptive figures [3] (2073 Bhadra, Q4a)
5. How does the GPS work? [4] (2078 Chaitra, Q4b)
6. Describe briefly on how GPS works? [Not specified] (2075 Bhadra, Q10b)
7. Illustratively explain how position is determined using GPS [Part of 6] (2073 Magh, Q8)
8. Explain how GPS can find position of a point [6] (2072 Magh, Q11)
9. Write short notes on: How the time is maintained in GPS [2] (2074 Bhadra, Q6ii)
10. How time is maintained in GNSS? [Not specified] (2075 Magh, Q4d)

10.3 DGPS

- **Theory Questions**

1. Explain Principal of DGPS [2] (2071 Bhadra, Q8c)
2. Why do we need DGPS even with the development of sophisticated GPS systems? [3] (2078 Chaitra, Q4b)

10.4 Errors in GPS

- **Theory Questions**

1. What types of error can occur during a GPS measurement? [Part of 6] (2073 Magh, Q8)
2. Explain different types of errors in GPS [2] (2071 Bhadra, Q8a)
3. Explain the errors that may occur during the time of GPS observations [2] (2071 Magh, Q5a)
4. Explain the possible errors that might occur while using GPS. How can accuracy of GPS be increased? [3] (2073 Bhadra, Q4b)
5. What are the sources of error in GPS measurements? How can the accuracy be increased? [3] (2076 Bhadra, Q4b)
6. Explain sources of errors in GPS and their corrections [8] (2072 Ashwin, Q5a)
7. How can you use GNSS (GPS) for navigation and survey? Explain multipath error in GNSS [3+3] (2079 Chaitra, Q4a)

10.5 Application

- **Theory Questions**

1. Write down the applications of GNSS [Not specified] (2075 Magh, Q4e)
2. Discuss the advantages and limitations of using smartphone apps for GNSS-based field survey [3] (2080 Chaitra, Q6)

Chapter 11: Introduction to Remote Sensing [4 Hours]

11.1 Concept of Remote Sensing

- **Theory Questions**

1. What is remote sensing? Explain the significance of spatial resolution and temporal resolution in remote sensing [2+4] (2071 Magh, Q5b)
2. Write short notes: Active and passive Remote sensing [2] (2074 Bhadra, Q6v)
3. Explain Active and Passive Remote Sensing [4] (2074 Magh, Q4c)
4. Explain sensors and platforms [2] (2071 Bhadra, Q9a)
5. Explain the following terms with example:

   a. Sensor

   b. Platform

   [2+2] (2072 Magh, Q11)
6. What do you mean by multi-spectral, multi-temporal and multi-sensor? [Not specified] (2075 Bhadra, Q11a)

11.2 Electromagnetic Spectrum and Spectral Signature of Different Landuse

- **Theory Questions**

1. Explain spectral signature of object [2] (2071 Bhadra, Q9b)
2. Explain the following terms with example: (c) Spectral signature [2] (2072 Magh, Q12c)
3. Explain the spectral signature of an object [Not specified] (2075 Magh, Q5e)
4. Explain about the spectral signatures in Remote Sensing. Why are they important? [4] (2078 Chaitra, Q9a)
5. How objects are differentiated in remote sensing by their spectral signatures, explain with illustration? [Part of 5] (2073 Magh, Q9a)
6. Write down spectral signatures of different land uses. [5] (2072 Ashwin, Q4c)
7. Explain spectral signature of different land use with example and application of Remote sensing in civil engineering [3+3] (2073 Bhadra, Q9)
8. Discuss the importance of understanding wavelength bands in remote sensing applications [4] (2080 Chaitra, Q5a)

11.3 Introduction to Different Windows

- **Theory Questions**

1. What is a band combination? Explain the uses of TRUE COLOR, FALSE COLOR, and SWIR band combinations with reference to LANDSAT images [4] (2078 Chaitra, Q9b)
2. What band combination used in Landsat 8 imagery to represent the natural colour and colour infrared (for vegetation)? [2] (2073 Magh, Q9b)
3. How do you create false color composite image? What is its importance? [Not specified] (2075 Bhadra, Q11b)

11.4 Resolutions in RS

- **Theory Questions**

1. Explain Resolutions in Remote Sensing. [4] (2074 Magh, Q4d)
2. What are different types of resolutions in remote sensing? [3] (2079 Chaitra, Q9a)
3. Explain different types of resolutions in RS [3] (2076 Bhadra, Q9b)
4. Explain the different types of resolution in remote sensing [Part of 5] (2073 Magh, Q9a)
5. Briefly explain spatial, spectral resolution, radiometric resolution, and temporal resolution [4] (2080 Chaitra, Q5b)
6. Write down steps for image processing [2] (2071 Bhadra, Q9c)
7. Write down the different image processing steps in remote sensing [Not specified] (2075 Magh, Q5c)
8. What is the meaning of 'training data' in supervised classification? [4] (2080 Chaitra, Q5c)
9. Write short notes: Supervised and unsupervised classification. [2] (2074 Bhadra, Q6vi)
10. What are the different types of resolutions in remote sensing imagery and their implications for data analysis? (2081 Chaitra)
11. Explain the meaning of each of the following terms related to remote sensing: a) Spectral signature b) Resolution of remote sensing data c) False color image d) Image classification (2082 Shrawan)

11.5 Application of Remote Sensing

- **Theory Questions**

1. What are the application of remote sensing? Explain some of the recent developments in the field of remote sensing [4] (2076 Bhadra, Q9a)
2. (Also covered under 11.2 in 2073 Bhadra, Q9.)
3. How can you find the change in forest cover of a specific location (such as a given city or province) over time using remote sensing? Describe the GIS and remote sensing analysis steps and the outputs. (2081 Chaitra)

Chapter 12: Making Maps [3 Hours]

12.1 Map Functions in GIS

- **Theory Questions**
  - No direct questions asked.

12.2 Map Design

- **Theory Questions**

1. Explain map design, layout, and visual hierarchy [Not specified] (2072 Magh, Q13)
2. What are the visual variables in map designing? [2] (2079 Chaitra, Q9b)
3. Explain about visual variables in map designing [3] (2076 Bhadra, Q10a)
4. Explain about visual variables in map designing. [3] (2073 Bhadra, Q10b)
5. How can colors and sizes of symbols be used to efficiently convey a map to the users? Explain with examples. (2082 Shrawan)

12.3 Map Elements

- **Theory Questions**

1. What is a map layout? List some of the important marginal information in a map. In which formats could the GIS maps be exported? [2+2+2] (2071 Magh, Q4b)
2. Briefly explain the map elements [6] (2073 Magh, Q10)
3. Explain map elements used in GIS mapping. [6] (2072 Ashwin, Q3b)
4. What are the elements to be included in a map layout? Explain with a sample layout [Not specified] (2075 Bhadra, Q12)
5. Prepare a standard layout of a map with all map elements [4] (2074 Bhadra, Q1a)
6. Prepare a standard layout for map printing [Not specified] (2075 Magh, Q1d)
7. Explain thematic map. Where and why do we design choropleth maps? [4] (2078 Chaitra, Q10)
8. Discuss the key considerations involved in designing a thematic map for effective communication. Explain the importance of selecting appropriate map elements (e.g., scale, legend, symbols) for a specific audience. (2081 Chaitra)

12.4 Choosing a Map Type

- **Theory Questions**

1. Explain about the types of maps [3] (2076 Bhadra, Q10b)
2. Explain the meaning of thematic map. (2082 Shrawan)

12.5 Exporting Map in Different Format

- **Theory Questions**
  - (Covered under 12.3 in 2071 Magh, Q4b.)

12.6 Printing a Map

- **Theory Questions**
  - No direct questions asked.